

Origin-Destination Matrix Estimation from Traffic Measurements

Experiments report — Tehran, District 2 (1,897 road links) · SUMO + Graph Neural Networks

Abstract. We study the inverse traffic problem: recovering an origin-destination (OD) trip matrix from road-level traffic measurements. Across four experiments we establish one central result — **the OD matrix is only as recoverable as the measurement is informative**, and the bottleneck is the *data*, not the model. Aggregate link counts leave the OD under-determined (cell-correlation 0.14 for an optimal linear inverse); **junction turn counts** make the zone marginals recoverable (correlation ≈ 0.90) and, combined with a doubly-constrained gravity reconstruction, yield **OD cell-correlation 0.78** at 36 zones. Real-world **travel-time** data (Neshan) is the weakest signal of all (ceiling 0.06): it can be used to *calibrate* a SUMO scenario that reproduces observed conditions, but it cannot identify the true OD. We report results at 36, 9 and 13 zones, a real-data calibration pipeline, and the engineering defects found along the way.

1. Objective and the inverse problem

The forward pipeline (`od_generator/`, generalised in `src/odpipe/`) samples an OD matrix, routes it through SUMO, and records per-edge traffic. We build the **inverse**:

per-edge SUMO measurements $\rightarrow$ **GNN** $\rightarrow$ OD matrix (origin $\times$ destination over TAZ zones).

This is fundamentally **ill-posed**: many different OD matrices produce nearly the same link measurements. Every result below follows from how much of the OD a given measurement actually constrains.

2. Method

The recipe that works, used identically in every experiment:

1. **Road line-graph** — nodes = the 1,897 road edges (features: length, lanes, speed limit, plus measured flow / speed / density / travel time and turn aggregates); edges = downstream connectivity; **edge weights = junction turn counts**.
2. **Predict the recoverable quantity** — a GNN (`ODMarginalGNN`, `GraphConv`) pools the graph into per-zone vectors and predicts **production/attraction marginals**, trained with a per-zone *standardised MSE* (a $\log_{1p}$ +Huber loss under-disperses the marginals and collapses the reconstruction).
3. **Reconstruct the matrix** — doubly-constrained gravity / **Furness (IPF)** from the predicted marginals with a distance-decay deterrence.

Splits are stratified by demand tier so every split covers all demand regimes. Metrics: marginal correlation, OD cell-correlation (pooled, off-diagonal), total-flow error, and **GEH** on re-simulated link flows.

3. Experiment 1 — What measurement identifies the OD? (the central result)

We fixed the model class (ridge) and varied only the *input signal*, on synthetic data where the truth is known, under a realistic 77 % edge coverage.

input signal	zone-marginal corr	reconstructed OD cell-corr
static only (no traffic)	0.00	0.02
travel time (what Neshan provides)	0.25	0.06
speed	0.24	0.05
link flow (counts)	0.36	0.14
junction turn counts (node-aggregated)	0.46	0.19
all dynamic features	0.50	0.20
<i>ceiling: gravity from TRUE marginals</i>	1.00	0.87

Two conclusions drive the whole project. **(a)** No model — GNN, VAE or ridge — can recover the OD from travel time alone; the information is not there. **(b)** The gravity reconstruction from *true* marginals reaches 0.87, so **predicting marginals + gravity is near-optimal**; the remaining task is to predict marginals well, which turn counts enable.

4. Experiment 2 — 36 zones (turn counts + marginal GNN)

Re-simulating each scenario to capture junction turn counts, and feeding them as GNN edge weights, lifts marginal correlation from ≈ 0.5 to ≈ 0.90 .

approach (36 zones, test n=750)	OD cell-corr	total-flow err	RMSE
direct OD-cell regression (baseline)	0.06	0.44	6.78
marginal GNN + gravity, link counts	0.435	0.49	6.78
marginal GNN + gravity, turn counts	0.784	0.132	4.08

Marginals: production $r = \mathbf{0.904}$, attraction $r = \mathbf{0.916}$. The direct-regression baseline confirms the under-determination; the turn-count model is a 13× improvement in cell-correlation over it.

5. Experiment 3 — 9 zones, and 36 vs 9

The same recipe re-run at a coarser resolution (TAZ grid width 3000 m $\rightarrow$ 9 zones, 2,000 samples).

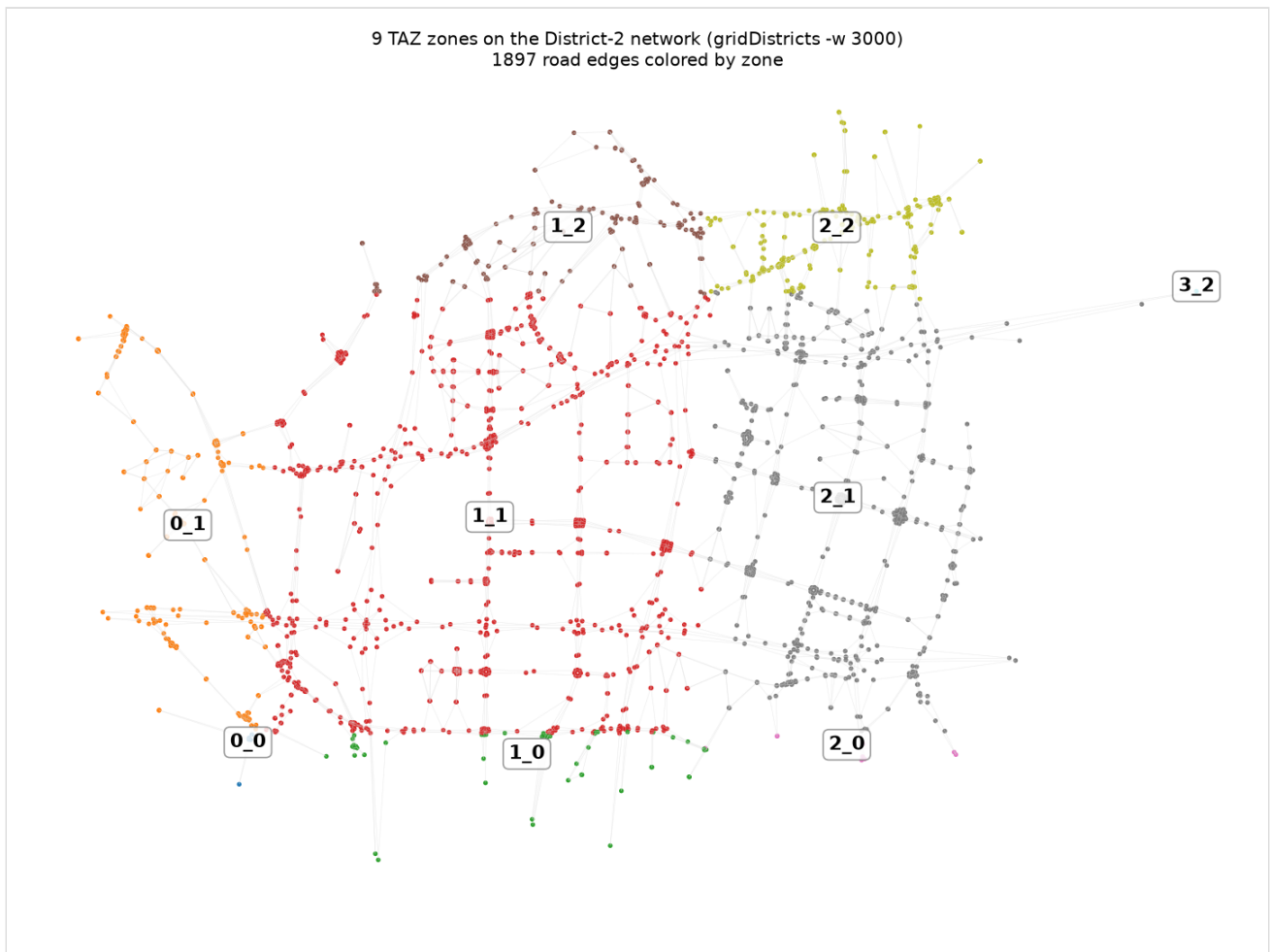


Figure 1 — The 9 TAZ zones on the District-2 network. Note the imbalance: the central zones dominate while edge cells are tiny — a recurring source of noisy marginals.

metric	36 zones	9 zones
OD cells (off-diagonal)	1,260	72
training samples	3,500	2,000
production / attraction corr	0.904 / 0.916	0.816 / 0.800
OD cell-corr	0.784	0.660
total-flow error	0.132	0.186
gravity ceiling	~0.86	0.884
GEH < 5 (link flows)	not scored	94.4 %

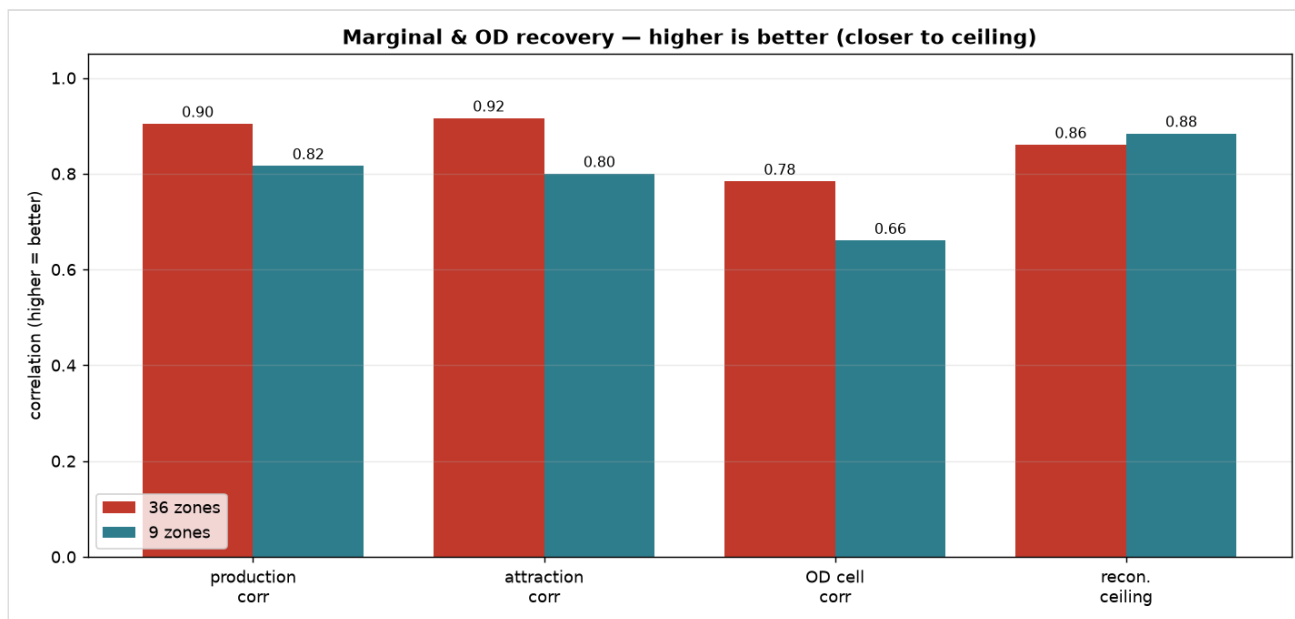


Figure 2 — Marginal and OD recovery at both resolutions, against the gravity ceiling. Both models sit close to their ceiling: the limit is information, not capacity.

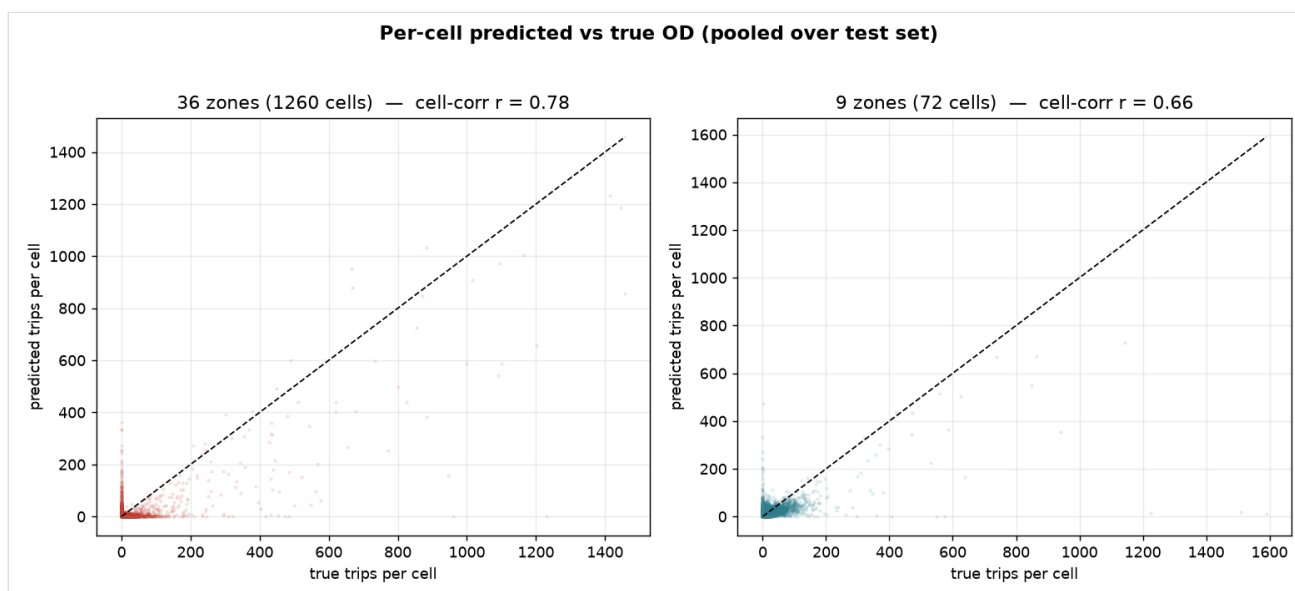


Figure 3 — Per-cell predicted vs true OD, pooled over the test set.

Finer is not harder here. The 36-zone model scores *higher* on every correlation metric despite having 17× more cells — largely because it was trained on 1.75× more data, and because coarse grids still contain tiny, near-empty zones that add noise. Conversely the 9-zone OD is the most **simulation-faithful**: re-simulating it reproduces observed link flows with GEH < 5 on 94 % of links. Raw RMSE is *not* comparable across resolutions (the same trips packed into 72 vs 1,260 cells make 9-zone per-cell values ~6× larger).

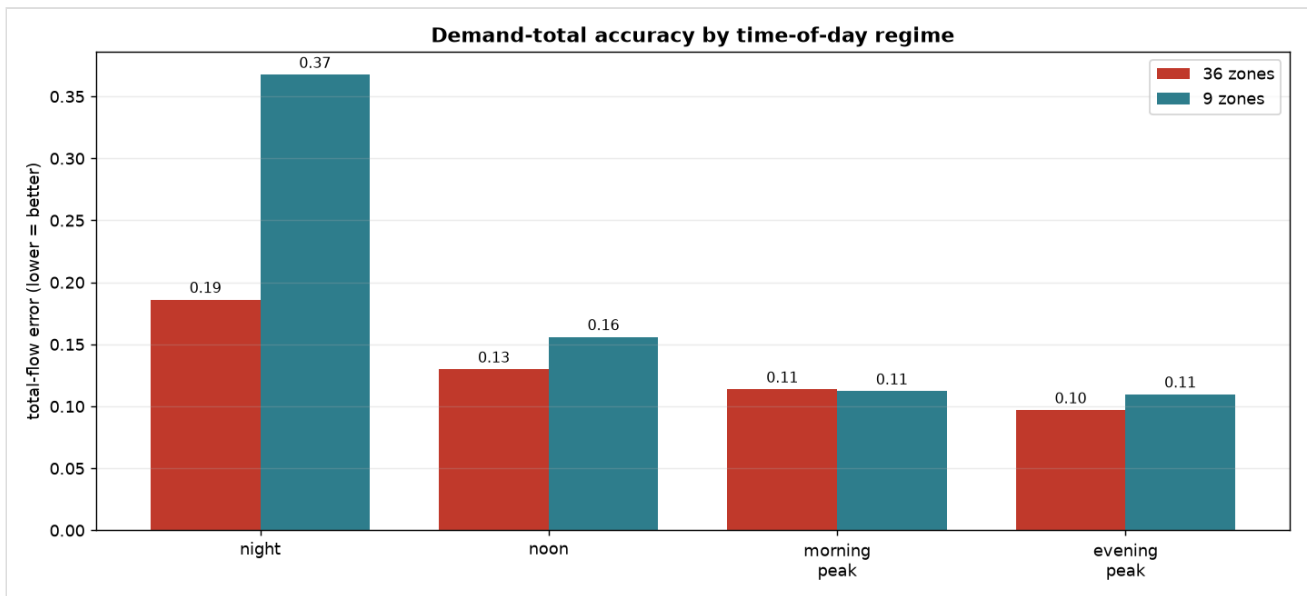


Figure 4 — Accuracy by demand regime. Both improve with demand; low-demand (night) is the hardest, where few trips give little signal.

6. Experiment 4 — General pipeline: 13 zones at high demand (mesoscopic)

To test generality we rebuilt the pipeline config-driven (`src/odpipe/`, nothing hard-coded to a zone count) and ran it at **TAZ width 2,000 m → 13 zones** with a **high demand range of 2,000-30,000 trips/hour**, simulated with SUMO's **mesoscopic** model (~2 s/sample — microsimulation is impractical at this demand).

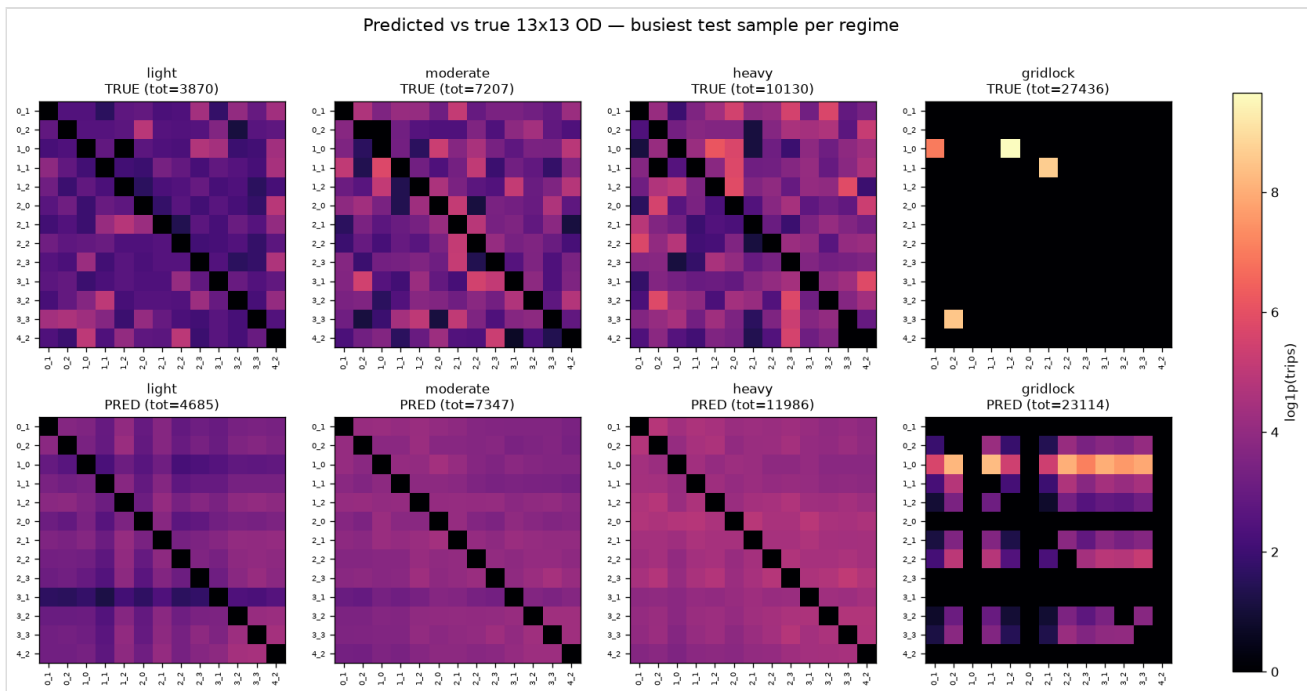


Figure 5 — Predicted vs true 13×13 OD, busiest test sample per demand tier. The blank rows/columns in the predictions are the three tiny zones (2-3 edges each) produced by the 2,000 m grid: they carry almost no traffic, so their marginals are unlearnable.

13 zones, high demand (smoke run: 160 samples, test n=24)	value
production / attraction corr	0.749 / 0.390
OD cell-corr	0.164
total-flow error	0.270
gravity ceiling	0.825
GEH < 5	48.3 %

These are **shakedown numbers from a deliberately tiny dataset** (160 samples, to validate the plumbing) and should not be compared with the fully-trained experiments above. Two real lessons do carry: the **grid width matters more than the zone count** — a width that produces near-empty zones (attraction $r = 0.39$ here) caps achievable accuracy; and **mesoscopic simulation is what makes high-demand study feasible** at all.

7. Experiment 5 — Real data: OD calibration from Neshan travel times

The operational goal is a SUMO scenario that reproduces *observed* traffic so that road changes can be simulated. The only real observable is **per-segment travel time** — by Experiment 1, the weakest signal (ceiling 0.06). We therefore do **not** learn the OD; we **calibrate** it with SUMO in the loop.

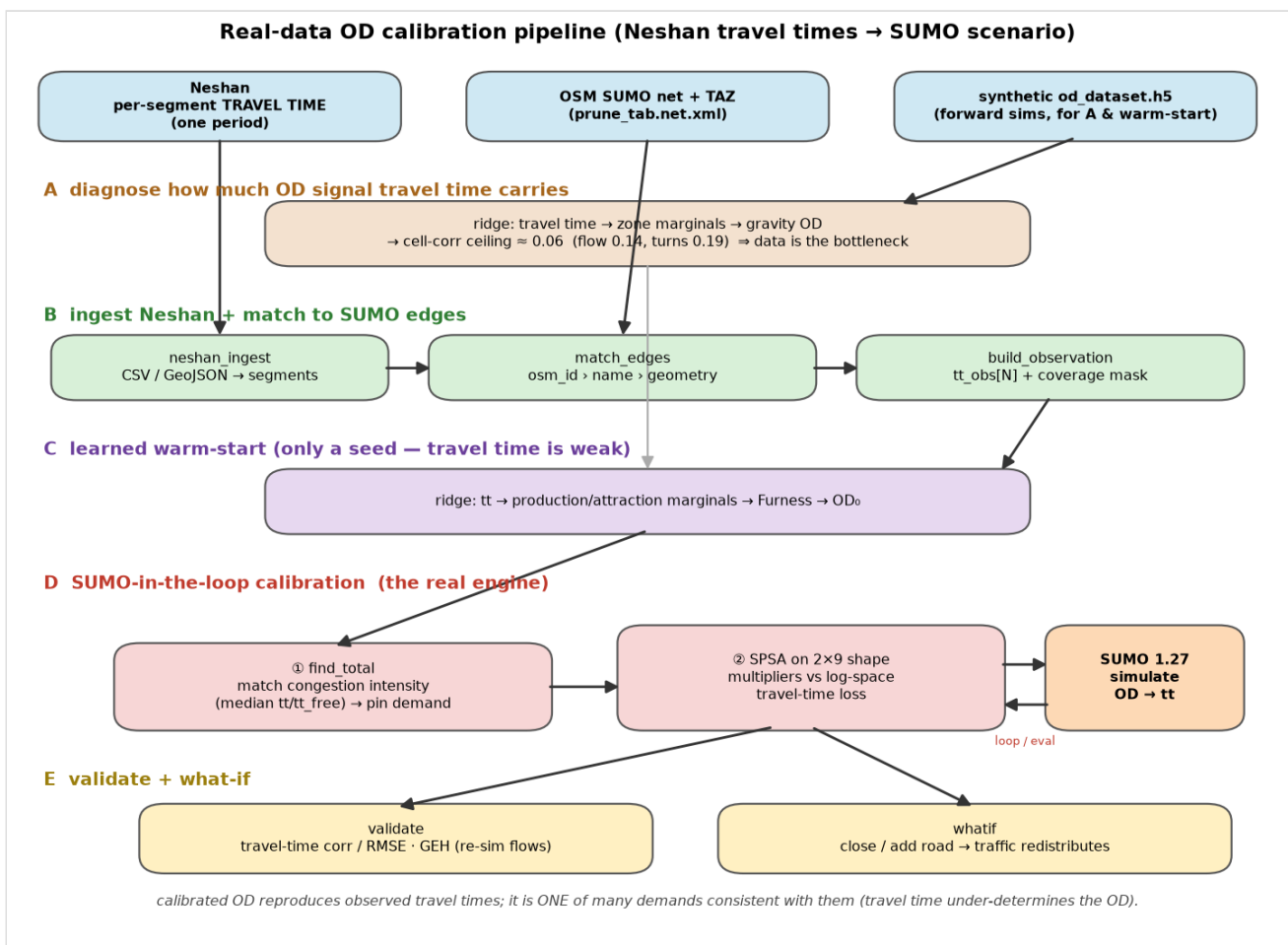


Figure 6 — The real-data calibration pipeline. A ridge model gives only a warm start; the engine is a SUMO-in-the-loop search that matches observed travel times.

The demand is split into the parts travel time can inform: **total demand** is pinned by matching the overall congestion intensity, then **SPSA** (gradient-free) optimises the spatial shape against a log-space travel-time loss on the observed edges.

On a self-consistent benchmark (a known 1,906-trip evening-peak OD, 398 observed edges):

	warm start	calibrated
travel-time correlation	0.677	0.722
travel-time log-RMSE	1.452	1.185
observed edges actually reproduced	58 %	77 %
SPSA loss (30 iterations)	2.110	1.412
recovered total demand	—	2,322 vs 1,906 true (+22 %)
OD cell-corr vs truth	—	0.035

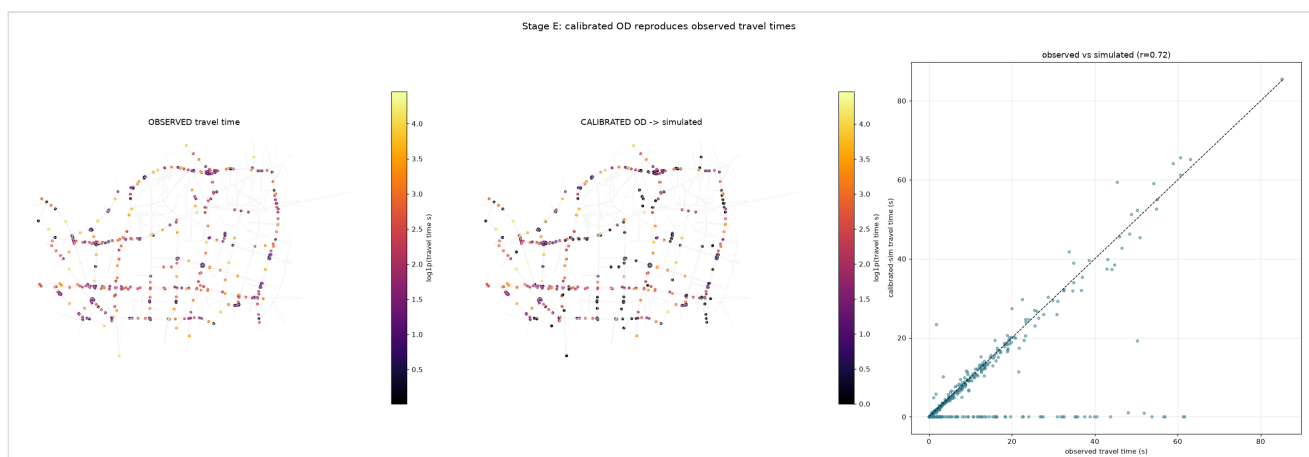


Figure 7 — Observed vs simulated travel times after calibration. Points on the horizontal axis are observed edges the calibrated demand leaves empty — a direct symptom of the non-uniqueness.

The honest result. We can recover the *demand magnitude* to ~20 % and reproduce observed travel times, **but not the OD itself** (cell-corr 0.035) — exactly as Experiment 1 predicts. Travel time lets you fit *either* the demand level *or* the per-edge pattern; forcing a tighter per-edge match (travel-time corr 0.90) requires inflating demand 3x. The calibrated matrix is therefore **one of many** demands consistent with today's data.

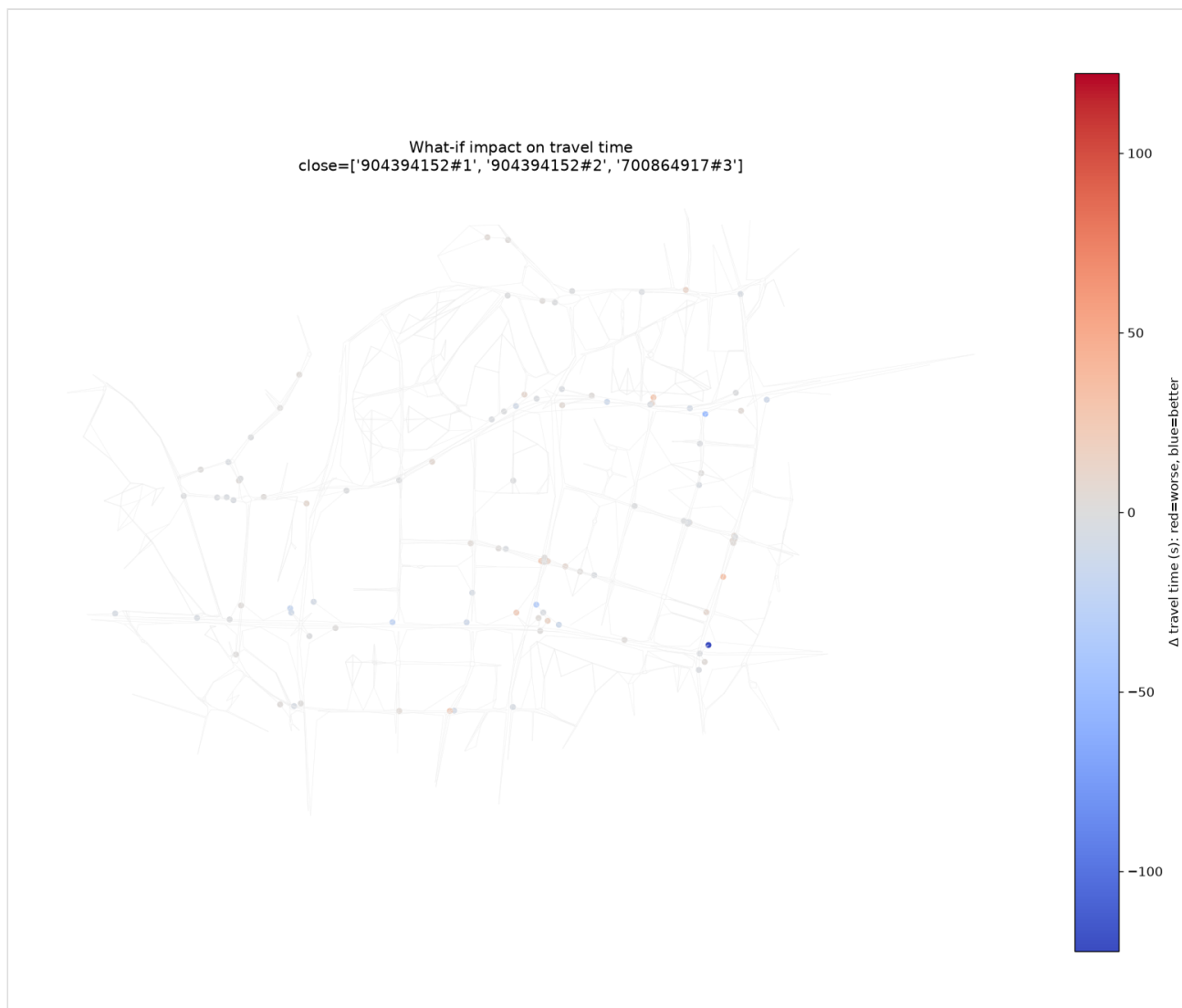


Figure 8 — What-if: closing a three-edge corridor with the calibrated demand held fixed. Travel time changes on 104 edges — relieved links in blue, detour links in red. This is the end goal working; its reliability is highest for moderate, local changes.

8. Engineering findings

Defects found and fixed — each one materially affected results:

finding	impact
Route files must be globally sorted by departure. SUMO streams route files and <i>silently drops</i> any vehicle departing before the previous one. Trips were written per OD pair, so most were discarded.	Simulations ran on a fraction of the intended demand (sparse traffic, under-counted congestion, biased calibration). Fixed by sorting all trips by depart time.
Race in parallel generation. Workers shared a temp directory keyed by <code>idx % n_workers</code> , so a freed worker could overwrite a running one's trip files.	~40 % of samples failed. Fixed with a unique directory per sample → 0 failures.
SUMO version mismatch. System SUMO 1.18 cannot parse this network (built with 1.26).	Pinned to the <code>eclipse-sumo 1.27</code> wheel and resolve binaries from it.

finding	impact
Marginal loss choice. log1p + Huber under-disperses the marginals (std-ratio 0.35), collapsing the Furness reconstruction.	Per-zone standardised MSE restores dispersion; cell-corr 0.44 → 0.78.
Long processes are killed (~45 s).	All training and calibration are checkpointed and resumable (- -max-seconds).

9. Conclusions

1. **The measurement, not the model, is the bottleneck.** Ranked by OD identifiability: turn counts > link counts > speed $\approx$ travel time. Adding model capacity or synthetic data cannot cross this ceiling.
2. **Predict marginals, reconstruct with gravity.** The full OD is under-determined; the marginals are not. This reframing plus turn counts takes cell-correlation from 0.06 to **0.78**.
3. **Choose resolution by purpose.** 36 zones for spatial detail and the best correlations; 9 zones for a compact, simulation-faithful OD (GEH < 5 on 94 % of links). Avoid grid widths that create near-empty zones.
4. **For the real network, calibrate — don't learn.** Travel time cannot identify the OD, but SUMO-in-the-loop calibration produces a scenario that reproduces observed conditions and supports what-if road changes, with quantified uncertainty.

Recommended next step: obtain any *count-like* measurement (junction turn counts or link volumes) for the real network. That single change moves the real-data problem from the 0.06 regime to the 0.78 regime, and makes SUMO's own `routeSampler/cadyts` calibration directly applicable.

10. Reproducibility

All hyperparameters and paths live in `configs/*.yaml`; seeds are fixed; CUDA is auto-selected with CPU fallback.

```
conda activate traffic

# general pipeline (any zone count / demand range) – see README_odpipe.md
python -m src.odpipe.dataset          # TAZ -> graph -> OD samples (.pkl) ->
packs
python -m src.odpipe.train            # train + evaluate
python -m src.odpipe.infer --max-trips --sumo-gui --which both --meso
python -m src.odpipe.visualize

# real-data calibration from Neshan travel times – see README_realworld.md
python -m src.realworld.diagnose_traveltime # the identifiability table (Experiment 1)
python -m src.realworld.calibrate --max-seconds 300
python -m src.realworld.validate
python -m src.realworld.whatif --close-edges <e1>,<e2>
```